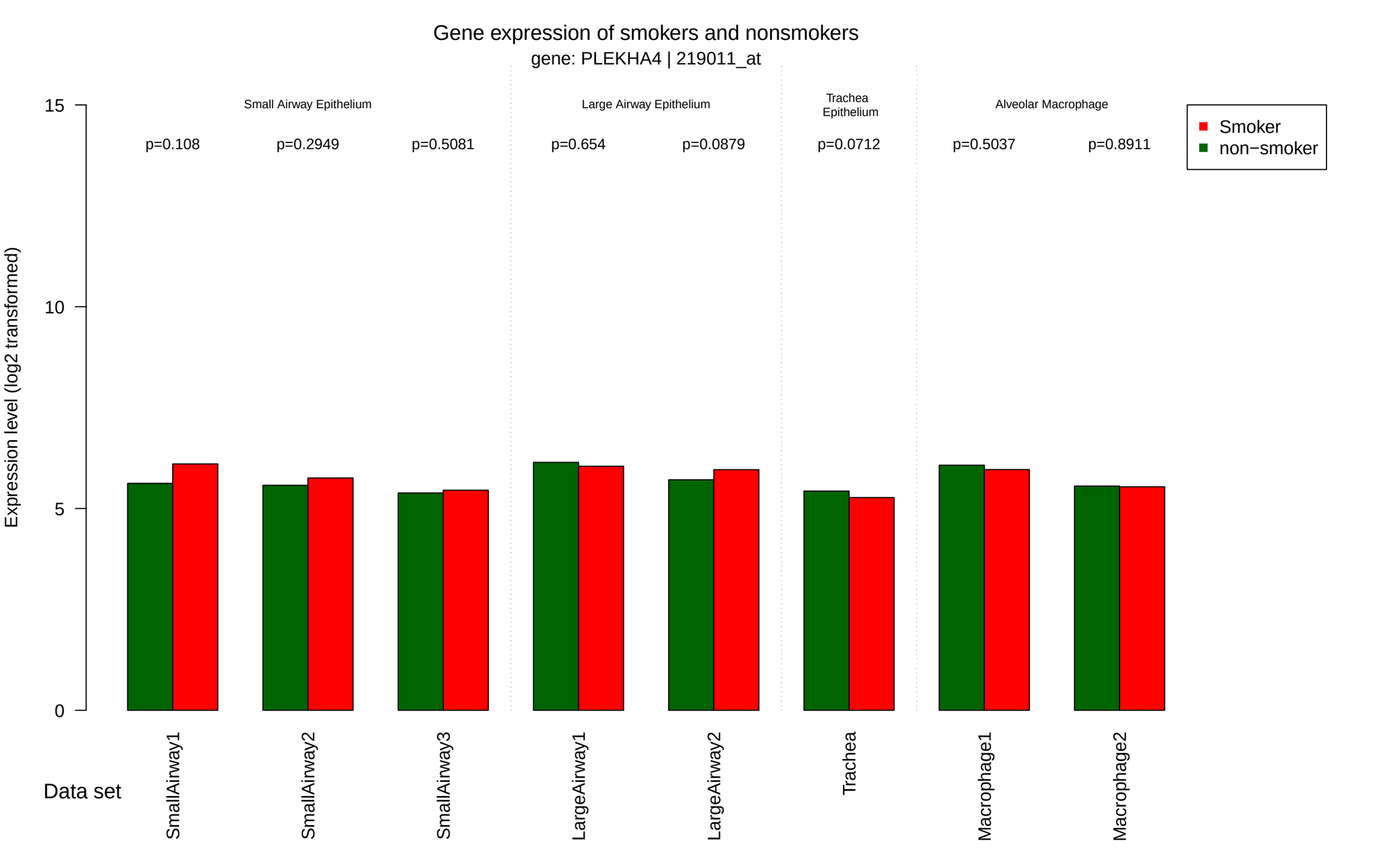
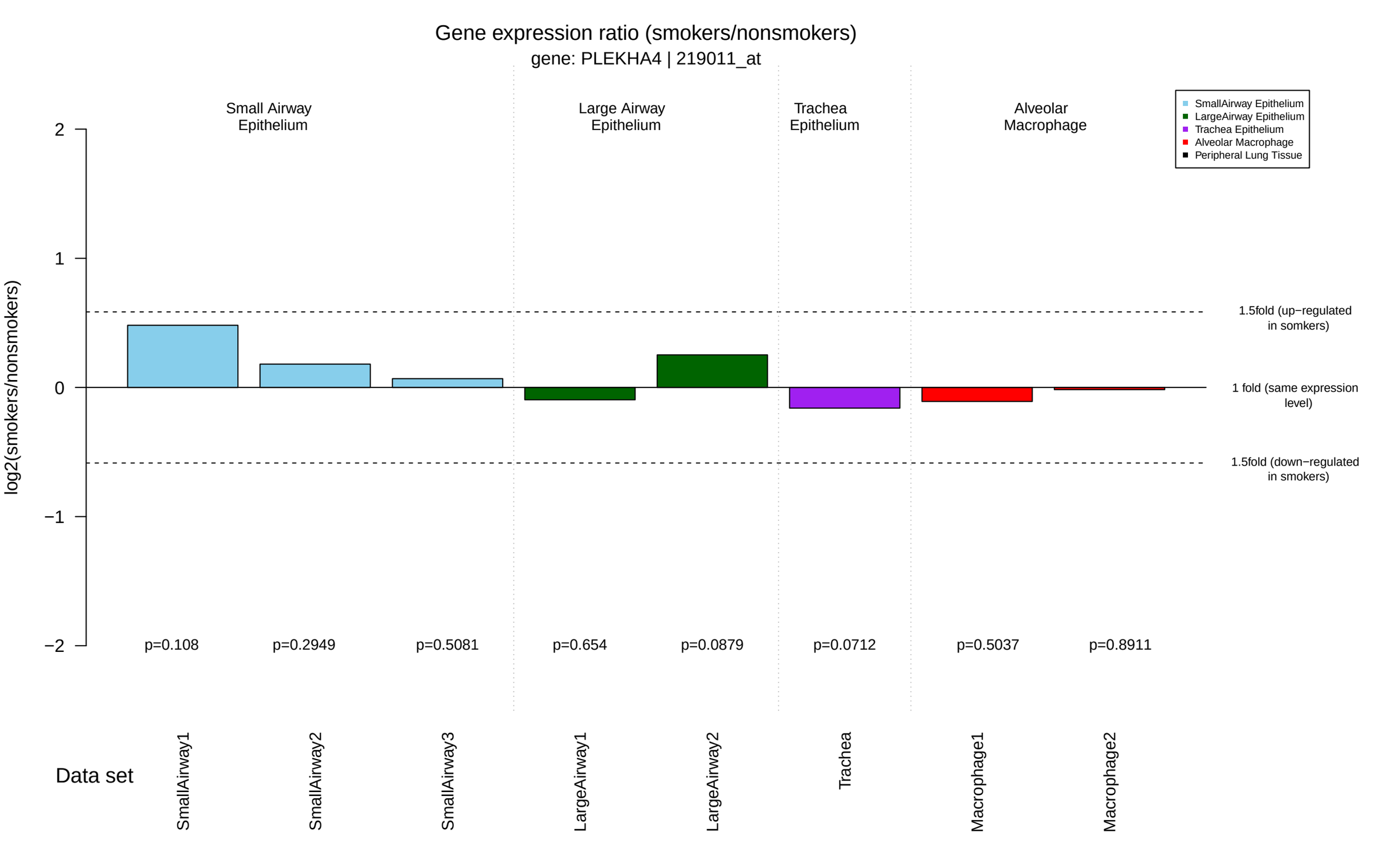
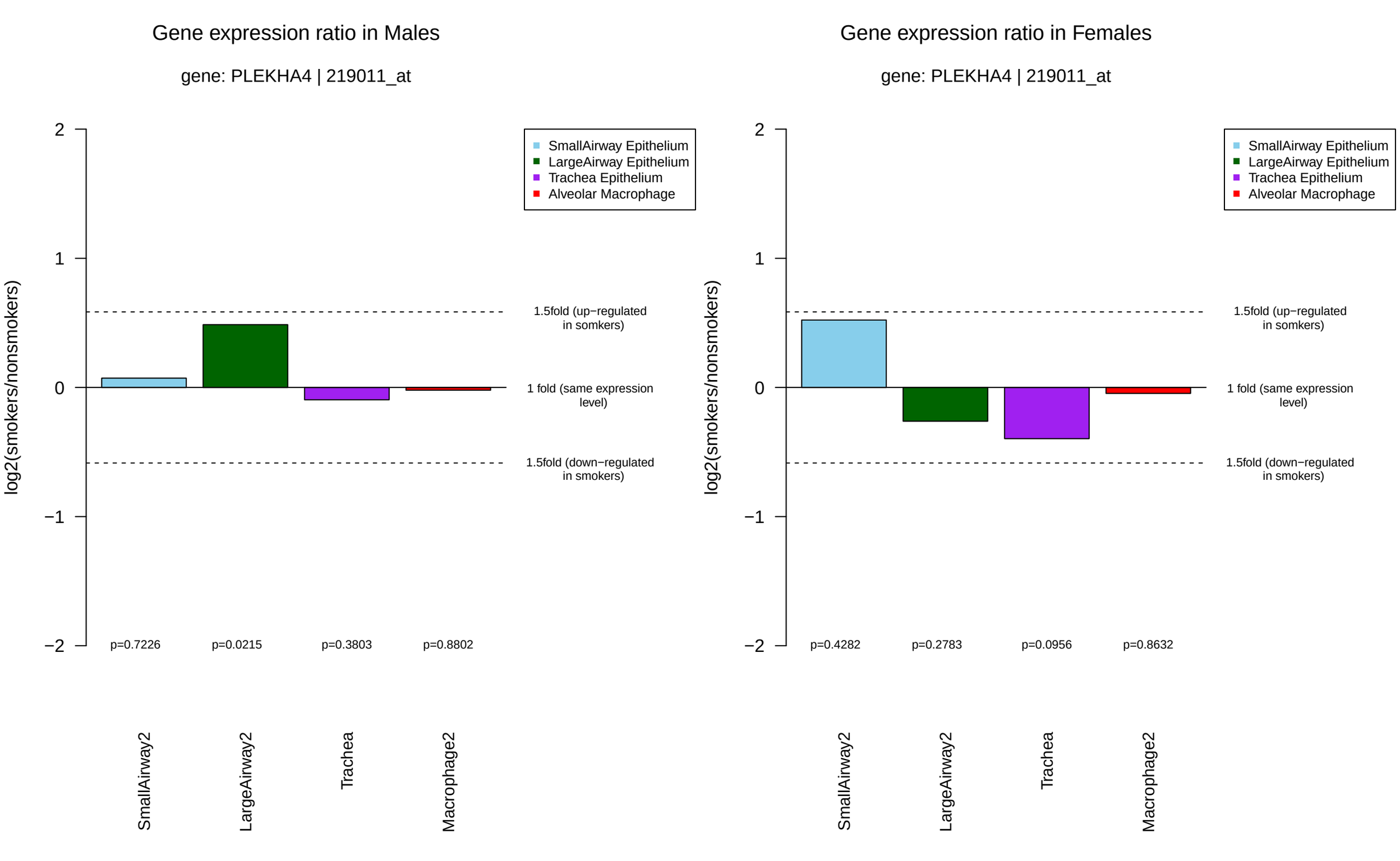
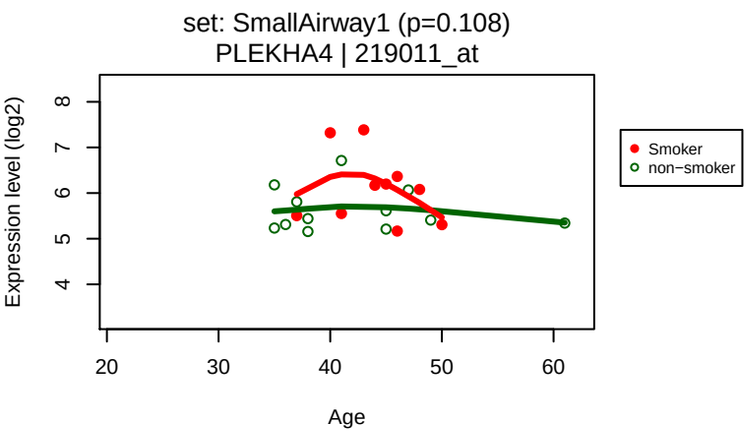
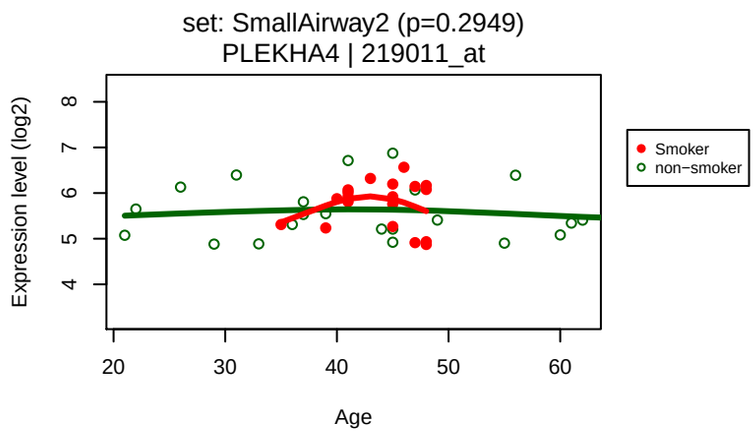
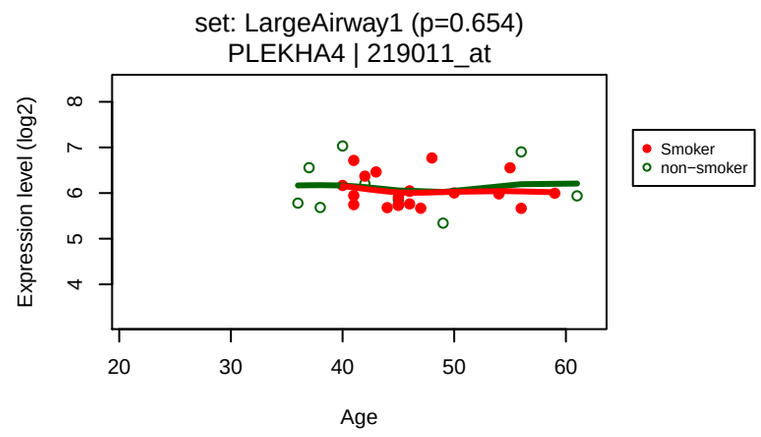

Gene expression of smokers and nonsmokers
gene: PLEKHA4 | 219011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.108
p=0.2949
p=0.5081
p=0.654
p=0.0879
p=0.0712
p=0.5037
p=0.8911
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

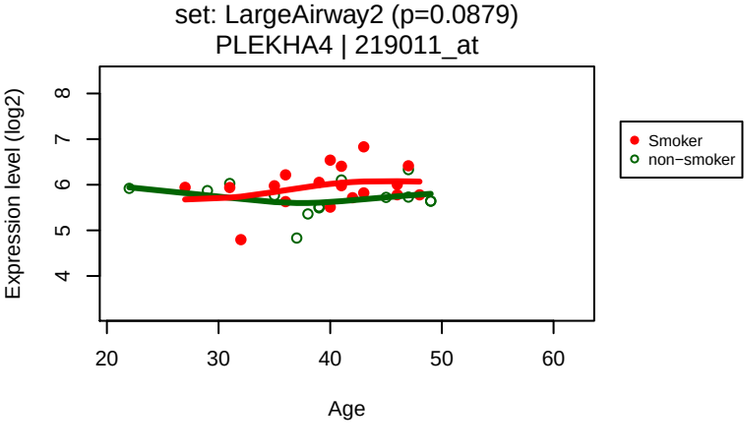

Gene expression ratio (smokers/nonsmokers)
gene: PLEKHA4 | 219011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.108
p=0.2949
p=0.5081
p=0.654
p=0.0879
p=0.0712
p=0.5037
p=0.8911
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

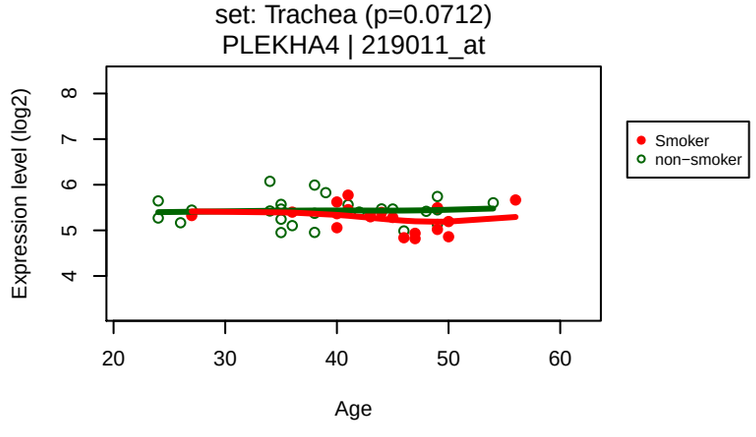

Gene expression ratio in Males
gene: PLEKHA4 | 219011_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7226
p=0.0215
p=0.3803
p=0.8802
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEKHA4 | 219011_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4282
p=0.2783
p=0.0956
p=0.8632
SmallAirway2
LargeAirway2
Trachea
Macrophage2

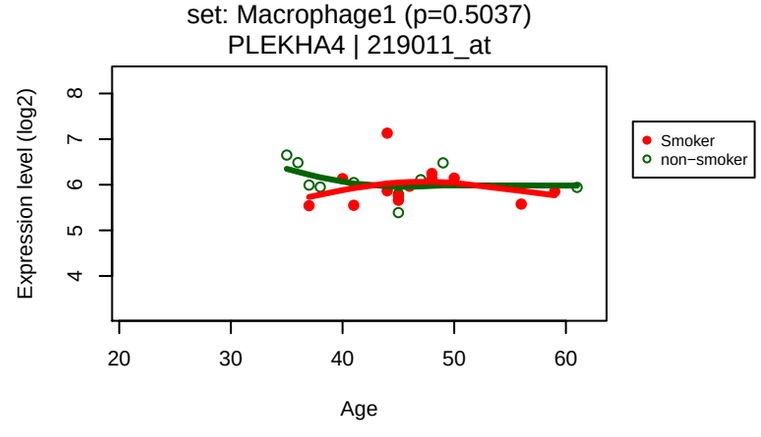

set: SmallAirway1 (p=0.108)
PLEKHA4 | 219011_at
Expression level (log2)
Age
Smoker
non−smoker

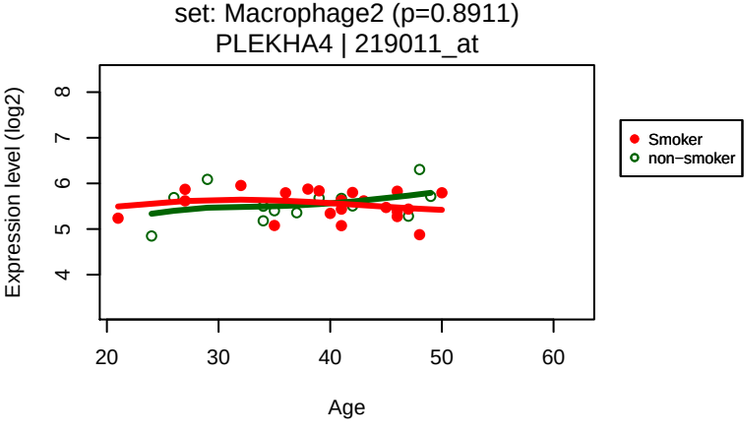

set: SmallAirway2 (p=0.2949)
PLEKHA4 | 219011_at
Expression level (log2)
Age
Smoker
non−smoker

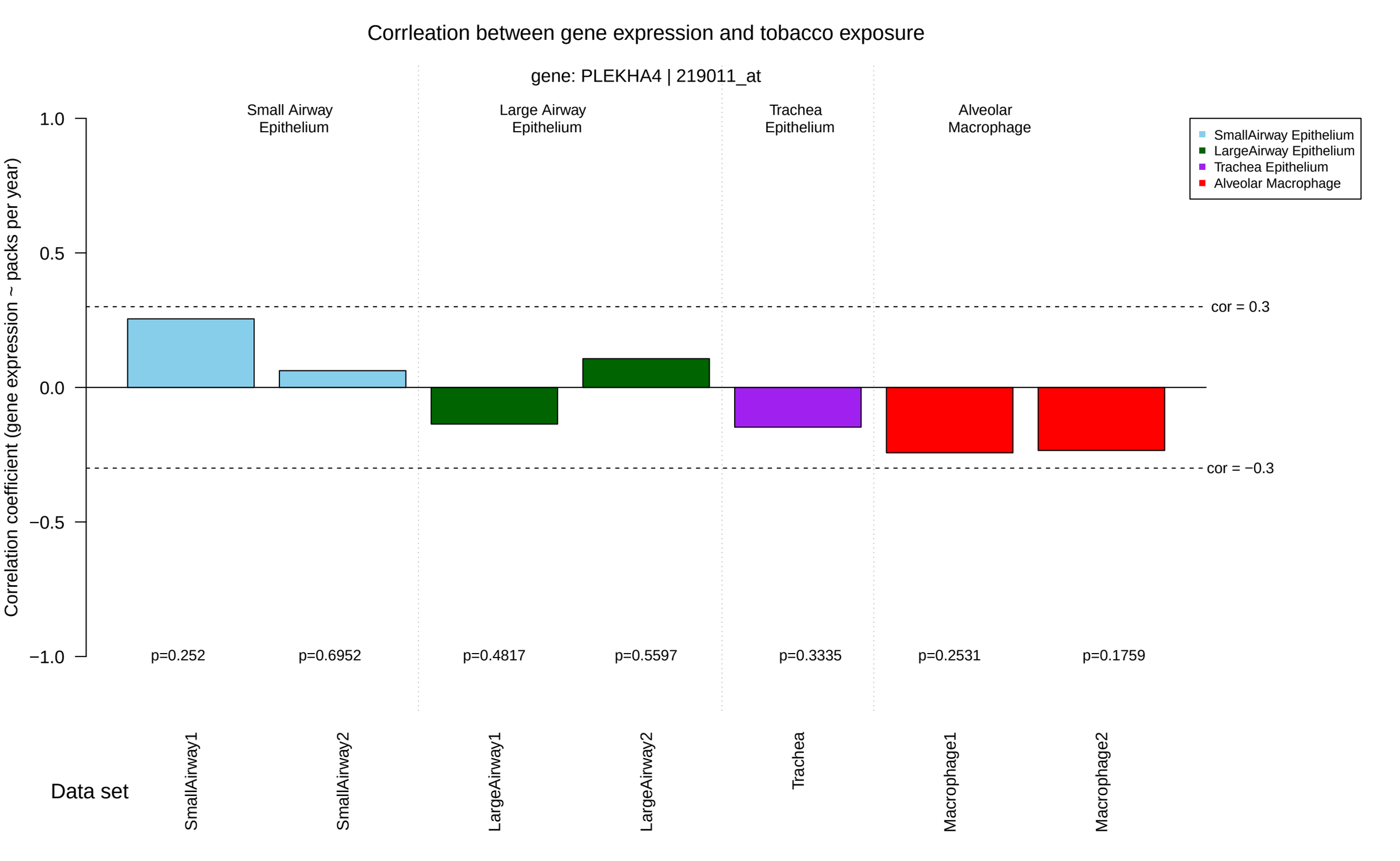

set: LargeAirway1 (p=0.654)
PLEKHA4 | 219011_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0879)
PLEKHA4 | 219011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0712)
PLEKHA4 | 219011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5037)
PLEKHA4 | 219011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8911)
PLEKHA4 | 219011_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEKHA4 | 219011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.252
p=0.6952
p=0.4817
p=0.5597
p=0.3335
p=0.2531
p=0.1759
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2